

Markup Typography on WebAssembly

Abstract

Modern document preparation systems demand both instant feedback and high typographical precision. While their compilation latency and complex dependencies hinder cloud-native interactive workflows. Typst introduces a new paradigm. In this work we present **moon-typst** a typographical compiler implemented in MoonBit. Its parser box layout and vector renderers can be measured separately with the checked-in benchmark.

Mathematical Foundation

The vertical spacing distribution model follows standard typographical baseline kinetics

$$E = mc^2$$

Furthermore inline formulations such as $f_x = \hat{x}^2 + 1$ and fraction derivations

$$\frac{a + b}{c + d}$$

maintain harmonic font metrics and balanced baseline alignments across heterogeneous output devices.

Architecture Overview

The system pipeline consists of three decoupled phases

- 1. Lexical Tokenization & Parsing Recursive descent parser yielding lossless AST nodes
- 2. Geometric Layout Engine Bounding box calculation font metric estimation and page break partitioning
- 3. Dual-Target Vector Renderer SVG 1.1 vector paths and semantic HTML5 DOM structures

Empirical Results

Run `moon bench benchmarks --target wasm-gc --release --deny` and record the host MoonBit version and raw results before making WebAssembly performance comparisons.